

Cole Plagens

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EDUCATION

University of Houston

Houston, TX

Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 3.64

Dec. 2025

- Relevant Coursework: Database Systems, Data Structures and Algorithms, Data Science I, Data Science and Statistical Learning, Software Design
- Dean's List Fall 2023 - Fall 2025

PROJECTS

Startup Operations Platform

Jan. 2025 – Present

- Architected and built a multi-tenant team operations platform supporting organization management, role-based access, project tracking, ticketing, analytics, and subscription workflows using React (TypeScript) and ASP.NET
- Containerized and deployed the system using Docker and AWS, integrated S3 for secure file storage, and configured a GitHub Actions CI/CD pipeline to automate testing, image builds, and cloud deployments, with the frontend hosted on Vercel

Volunteer Activity Application

Sept. 2025 – Nov. 2025

- Built a full-stack volunteer management application using the PERN stack, enabling secure user authentication, role-based access for admins and volunteers, and full event lifecycle management
- Implemented an automated volunteer–event matching system based on user profiles, skills, and availability, with notifications and participation tracking to streamline coordination and improve engagement, all powered by Supabase

Theme Park Management System

Jan. 2025 – Apr. 2025

- Developed a full-stack Theme Park Management System using React and ASP.NET with role-based authentication supporting ticket purchasing, ride browsing, and user account management
- Designed a relational MySQL database with complex queries and trigger-based constraints to enforce data integrity for ride status, maintenance workflows, and featured attractions
- Built responsive admin dashboards with CRUD operations and analytics reporting for revenue, ticket sales, maintenance costs, and ride performance, deployed via Azure and Vercel for scalable delivery

EXPERIENCE

CougarCS Tutor

Sept. 2025 – Dec. 2025

University of Houston

Houston, TX

- Debugged complex C/C++ implementations and guided students through algorithmic problem-solving and systems-level programming challenges
- Analyzed student codebases to identify logic errors, runtime issues, and inefficiencies, improving solution correctness and reinforcing best practices in debugging and optimization

Crew Member

July 2022 – Aug. 2023; May 2025 – Present

Firehouse Subs

Houston, TX

- Delivered customer service in a fast-paced environment while maintaining strict operational standards

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, C#, HTML/CSS, R

Frameworks, Libraries and Runtimes: React, Node.js, Express, ASP.NET, Next.js, Tailwind CSS

Cloud and DevOps: Git, GitHub, Docker, Supabase, AWS, Azure

Data Science / ML: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Jupyter Notebook, TensorFlow, PyTorch

Core Competencies: Distributed Systems, REST APIs, Database Design, Cloud Deployment, CI/CD